

Genetic engineering: change, test, decide

Teacher guide and worked answers

Describe genetic engineering and use evidence to weigh a specific benefit, a risk and an unanswered question.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C30; C31 (bridge)

Describe selective breeding & genetic engineering.

Second Week 3 exemplar: core Foundation-relevant genome modification (4.10) and initial evaluation (4.14). The Bt example also supports Biology-specific 4.12B. Prepares for C31 "Evaluate GM (research & digital slideshow; links RE)." It does not replace Week 4 independent research/slideshow. Enzyme/vector detail is optional Higher-tier stretch (4.11), never a core success gate.

Prior learning

A gene is a DNA section that can carry instructions affecting proteins/characteristics.

Selective breeding chooses which organisms reproduce across generations.

Natural selection can make inherited resistance more common in a population.

Success evidence

Describe genetic engineering as a direct alteration of an organism's DNA/genome.

Connect a genetic change to a useful product or trait without claiming a change of species.

Give an evidence-based provisional decision with one benefit, one risk and one further check.

Equipment

Shared trial brief and selected response page; pen or mini-whiteboard.

Offline HTML/PPTX; calculator optional for stretch comparison.

Preparation

Explain that the crop trial is a constructed sample, not a claim about a named product or UK approval.

Print the evidence page; no online reading is needed for this lesson.

Keep mechanism teaching brief and concrete. Optional enzyme detail is for pupils ready for Higher-tier material, outside the Foundation core gate.

Practical care

Paper and screen lesson; no live organisms, biological cultures or chemical work.

Ready to teach alternative

All evidence is supplied for offline use. Discuss, point, annotate or dictate instead of cutting cards or writing extended prose.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A crop makes a pest-control protein
2	Retrieval	2-5	Retrieve genes and selection
3	I do	5-9	What genetic engineering means
4	I do	9-12	Connect the change to the trait
5	I do	12-15	The product is the protein
6	I do	Within stage	Predict a risk using biology
7	We do	15-22	Trial board: what does the evidence show?
8	We do	Within stage	Spend your two next checks
9	Hinge	22-25	Which explanation is scientifically sound?
10	You do	25-36	Write a provisional recommendation
11	You do	Within stage	A recommendation is more than a vote
12	You do	Within stage	Optional Higher-tier extension
13	Review	36-39	Review a recommendation fairly
14	Review	Within stage	Carry a good question into Week 4
15	Exit	39-40	Explain, support, qualify

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A crop makes a pest-control protein

Take a prediction and preserve it for review. Say that genetic engineering describes a method; evidence must be about a specific application. Announce the independent product: a short trial recommendation for a grower, not a blanket vote on all GM organisms.

2 Retrieval Retrieve genes and selection

R1 a section of DNA with genetic information; many genes code for proteins. R2 which organisms reproduce, selecting for desired inherited characteristics. R3 yes, through differential survival/reproduction of inherited variants. Accept plain language then model the scientific term. No disclosure of personal medical or family information.

3 I do What genetic engineering means

Think aloud: 'Selective breeding chooses parents. Genetic engineering directly changes genetic information using laboratory technology. Both can alter characteristics, but the method differs.' Do not imply every DNA change alters phenotype or every engineered organism includes a foreign gene. The drawing uses symbolic segments, not real base sequences.

4 I do Connect the change to the trait

Point across DNA → protein → possible trait. Say: 'This gene carries information used to make a protein. The protein has a biological effect on certain target pests. I should not say the plant becomes a bacterium or kills every insect.' Use only a simplified model; expression depends on the cell and appropriate control information.

5 I do The product is the protein

Model the distinction: 'A useful instruction has changed what the cell can produce; it has not supplied a complete human genome. The medicine is a checked protein product.' This is a biological application, not medical guidance. Do not ask who uses insulin or why. Detailed manufacturing steps are outside this lesson.

6 I do Predict a risk using biology

Shares the preceding 3-minute I do stage. Briefly connect to natural selection: the pesticide protein does not teach insects to resist. Variants that already arise can be selected. Integrated pest-management and refuge strategies can delay resistance but are not a guarantee. This is a mechanism diagram, not observed data.

7 We do Trial board: what does the evidence show?

Spend 2 minutes reading the table, 2 minutes group/solo evidence verdict and 3 minutes additional-question selection on next slide. Constructed data show 22 fewer damaged plants per 100 and 1.0 t/ha greater mean yield. A provisional benefit at this site/season is supported. The table does not show the need for fewer sprays, food safety, effects on other species or long-term outcomes. Both groups received no target-insect spray.

8 We do Spend your two next checks

Shares 7-minute We do. Best choices from these four are repeat sites/seasons and resistance monitoring: they address generalisation and loss of efficacy. Crop colour does not answer the stated damage/yield/resistance question unless a new relevant mechanism is proposed; online popularity is not biological evidence. Other real evidence, including non-target effects and crop-specific safety assessment, remains necessary. Pupils can point to two choices and explain quietly.

9 Hinge Which explanation is scientifically sound?

Private response before feedback. Reteach A/D by revisiting the three types of DNA change; B by distinguishing gene information from whole-organism identity. C is the secure core definition.

A. Engineering always transfers DNA between species. - Adding another species' DNA is one method; DNA may also be changed or deleted.

B. An insulin-producing bacterium becomes human. - It remains a bacterium; it can make a useful protein.

C. Engineering alters DNA and may introduce a useful characteristic. - This describes the method without promising that every change works.

D. Engineering guarantees that all offspring are safer. - No blanket guarantee follows from the method; the actual organism and application require evidence.

10 You do Write a provisional recommendation

Independent product: a grower should understand what has been engineered, what this evidence supports and what remains to investigate. Choose Supported/Standard/Stretch, not all three. At minute 4 ask pupils to separate a measured benefit from a predicted risk; at minute 8 check their next step is specific.

11 You do A recommendation is more than a vote

Shares 11-minute independent stage. Reasoned recommendations can differ. Strong evidence supports continuing carefully controlled testing; simply asserting all GM is good/bad is insufficient. A supported oral response can meet all core success criteria.

12 You do Optional Higher-tier extension

Shares independent stage; do not teach this to all pupils or use it as a Foundation gate. The Stretch page supplies a conceptual explanation of a classical recombinant-DNA method including complementary sticky ends, not a lab protocol. Other engineering methods exist.

13 Review Review a recommendation fairly

Light Lundy loop: pupils choose written/oral/annotated response and an audience; the teacher responds to one practical suggested check and records a class next question for Week4. Avoid framing a pupil's preference as a scientific fact.

14 Review Carry a good question into Week 4

Shares the 3-minute review. This is a bridge to the SoW Week4 research/digital-slideshow evaluation, not its replacement. Record a question pupils can later research using checked sources, such as effectiveness in another climate or effects on non-target organisms.

15 Exit Explain, support, qualify

E1 direct alteration of DNA/genome by laboratory technology. E2 fewer damaged plants 30 → 8 per 100 or higher mean yield 6.9 → 7.9 t/ha in this constructed trial. E3 resistance may spread; monitor resistant pests/efficacy over generations. Accept other specific plausible risks paired with relevant tests, not ungrounded certainty.

Answers and assessment

R1-R3 retrieval

R1 a gene is a section of DNA containing genetic information; many code for proteins. R2 humans choose which organisms reproduce for desired inherited traits. R3 yes: inherited variants that survive and reproduce can become more common in later generations.

W1 / S2 / M3 data comparisons

Constructed trial: damage falls from 30 to 8 plants per 100, a difference of 22 plants per 100 (22 percentage points if expressed as percentages). Mean yield rises from 6.9 to 7.9 t/ha, a difference of 1.0 t/ha. These are means in one site/season, not universal outcomes. Optional relative damage reduction $22/30 \times 100 \approx 73\%$; do not confuse this with 22 percentage points.

W2 justified claim / S3 benefit

"This trial shows a promising benefit here" is supported: less target-pest damage and higher mean yield. "Success everywhere" or "proof that all GM crops are safe" is not. Neither group received target-insect spray, so this table does not directly measure a reduction in spraying. Future reduced spray use is a possible benefit requiring its own comparison.

W3 next checks

Repeat across sites/seasons tests whether the result generalises; monitor resistance tests whether pest susceptibility/effectiveness changes across generations. Crop colour is not relevant to the stated performance question without a further mechanism. Online likes measure popularity, not biological outcomes. These two checks still leave questions such as non-target effects and crop-specific safety assessment.

S1 / E1 definition

Genetic engineering directly alters an organism's DNA/genome using laboratory technology. A Bt gene can enable the crop to make a protein that harms particular insect pests. DNA may be added, changed or deleted; not all engineering transfers DNA between species.

S4 resistance mechanism

Pests contain inherited variation; those resistant to the protein may survive exposure and reproduce, passing on relevant alleles. Resistance can therefore become more common across generations. Do not accept that individual pests intentionally change because they need to survive.

S5 / M5 provisional recommendation

Model: continue controlled trials with resistance monitoring before making a wider-use recommendation. The initial reduction in damage is promising, but test matched crop groups over more sites/seasons and monitor pest survival/trait effectiveness. A consistent lack of yield benefit or increasing resistance would weaken the recommendation. Accept a reasoned alternative if it uses supplied evidence and states uncertainty.

M1 distinction

Selective breeding chooses parents with desired inherited traits, then breeds and selects across generations. Genetic engineering directly changes DNA using laboratory technologies. Both can influence inherited characteristics; the method distinguishes them.

M2 mechanism

The added gene provides information for production of a protein in suitable plant cells. That protein affects particular target pests and may reduce damage. The crop remains a plant; not every insect is affected, and the intended trait needs testing.

M4 risk and unknown

Risk: selection can increase inherited pest resistance, reducing future effectiveness. Unknowns: effects across settings, non-target species, food safety, long-term performance or social/economic consequences. Match the claim to what the table actually measures. Do not treat an unanswered question as proof of harm.

T1 two settings

First yield difference Bt–non-GM=+1.0 t/ha; second=–0.1 t/ha. First damage difference=22 fewer plants per100; second=1 fewer. The second setting weakens a blanket benefit claim and suggests context (including pest pressure) matters. It does not prove the first result false or that the crop is always ineffective; means, spread and trial replication matter.

T2 scope and investigation

The trials measure target-pest damage and yield, not composition/toxicity/allergenicity or every ecological interaction. A focused environmental investigation might compare abundance/survival of specified non-target organisms in matched replicated plots over time. A relevant crop-specific safety assessment would need different evidence. Do not infer safety/harm from yield alone.

T3 balanced evaluation

A defensible answer links the initial benefit to reduced target-pest damage, identifies resistance or a relevant possible environmental issue, notes missing repeated/appropriate evidence, and considers growers, consumers or ecosystems. A provisional decision must be conditional, not a universal judgement on the technique.

T4 optional Higher mechanism

Restriction enzyme cuts at a particular recognition sequence; complementary sticky ends can base-pair; ligase seals/joins the DNA backbone; a vector such as a plasmid carries DNA into suitable cells. Successful insertion alone does not guarantee expression, sufficient functional protein or the desired whole-organism effect, so suitable control information/conditions and checks are needed. This detail is optional Higher-tier content, not a Foundation assessment requirement.

Hinge A-D

C is correct. A wrongly says all methods transfer between species. B confuses one useful gene with complete organism identity. D makes an unsupported universal safety/offspring guarantee.

E2-E3 exit

E2 accept lower damage 30 → 8 per 100 or greater mean yield 6.9 → 7.9 t/ha, explicitly in the constructed trial. E3 accept resistance with monitoring across pest generations; accept another scientifically plausible specific risk with an appropriate focused test, not a blanket assertion.

L1-L2 reflection and Week4 bridge

Accept a concrete revision prompted by a question, such as naming the crop/trait, correcting a unit or identifying a missing comparison. Good Week4 questions are narrow and answerable, for example “How does this Bt trait affect a named non-target insect in a particular environment?” or “When does lower pest damage lead to a worthwhile yield gain?”

Teaching and assessment notes

40 minutes: opening², retrieval³, I do¹⁰, We do⁷, hinge³, independent¹¹, review³, exit¹. Zero-minute slides share their stage time.

Foundation core: genome modification, DNA/protein/trait reasoning and evidence-based evaluation. The optional Higher-tier box maps to detailed enzyme/ligase/sticky-end/vector content; do not gate the core outcome on it.

Week4 research and digital-slideshow work remains a separate SoW requirement. Carry forward pupils' questions; use sources checked for the particular application and distinguish evidence from value judgements.

All crop numbers are constructed sample data. The cells, proteins and selection diagrams are conceptual models, not practical instructions or exact molecular drawings.

Teacher model recommendation: “The first trial suggests less target-pest damage and higher yield at this site. I would continue controlled testing with resistance monitoring because a one-season result cannot establish long-term or wider outcomes.”

Assess success through an accurate definition, a causal link from genetic information to trait/product, one numerical evidence comparison and a justified next check. Oral and annotated work can meet the same standard.

If pupils infer all GM is good/bad, ask “Which organism, which change, which measurement?” If they confuse cells with insulin, point separately to producer and purified product. If they omit uncertainty, ask which table column could answer their claim.

No pupil medical or family disclosure is requested. Insulin is used only as an example of a protein product made through biotechnology; this lesson offers no advice on diagnosis or treatment.

Access and responsive teaching

Supported

Same scientific objective, with chunked cards, word bank, read-aloud access and short evidence stems. Accept accurately explained oral or annotated responses.

Standard

Use two evidence sources and a reasoned explanation independently.

Stretch

Evaluate limitations, explain a mechanism and transfer to unfamiliar evidence.

Regulation

Offer solo or partner working and a quiet response route. Preview the sequence; allow a brief reset and re-entry at the next numbered task. No compulsory performance or timed competition.

Misconceptions to check

All genetic engineering means moving a gene between species.

It can add, change or delete DNA; some methods do not add another species' DNA.

Check: Ask pupils to give two kinds of DNA change.

A bacterium given an insulin gene becomes human.

It remains a bacterium but can produce the coded protein under suitable conditions.

Check: Ask what is recovered and purified: the protein or a human organism?

Every GM crop has the same benefits and risks.

Assess the particular trait, crop, environment and evidence.

Check: Ask whether target-insect damage data measure food safety or effects on every species.

Pests become resistant because they need to.

Inherited variants differ in survival and reproduction; resistance can spread across generations.

Check: Ask what existed before exposure and what changed in later generations.

Word	Meaning
genome	An organism's complete set of genetic information.
genetic engineering	Using laboratory technologies to directly alter an organism's DNA.
gene expression	Using genetic information to make a functional product, such as a protein.
GM organism	An organism whose genetic information has been altered using genetic engineering.
Bt	Bacillus thuringiensis, a bacterium from which some pest-control protein genes originate.
provisional conclusion	A conclusion supported now, which can change when new evidence is added.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification, Topic 4

Used to check GCSE relevance; teacher-authored practice, not an official assessment. Checked 8 September 2026.

NHGRI: Genetic Engineering

Checked definition and the distinction between adding, altering and deleting DNA. Checked 8 September 2026.

NHGRI: Recombinant DNA Technology

Checked recombinant DNA and microbial production of useful proteins; used for conceptual explanation only. Checked 8 September 2026.

FDA: Science and History of GMOs and Other Food Modification Processes

Checked general Bt crop development and the need for testing; no US regulatory claims applied to UK law. Checked 8 September 2026.

US EPA: Insect Resistance Management for Bt Plant-Incorporated Protectants

Checked target-insect effects, potential crop benefits and resistance-management mechanisms; no current legal requirements or percentages taught. Checked 8 September 2026.